

Probabilistic Precipitation Nowcasting with Rectified Flow Transformers

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Abstract

Accurate weather forecasts are essential across various domains and are safety-critical in extreme weather conditions. Compared to simulation-based forecasting, data-driven approaches show greater efficiency, enabling short-term, high-resolution nowcasting. In particular, diffusion models proved effective in weather nowcasting due to their strong probabilistic foundation. However, existing methods rely on deterministic compression to reduce the complexity of high-dimensional weather data, limiting their ability to capture uncertainty in the decoding process. In this work, we introduce *FREUD*, a **F**rame-wise **e**ncoder and **U**nited **D**ecoder model based on rectified flow transformers for efficient compression of spatio-temporal weather data. Frame-wise encoding enables continuous forecast updates, while the unified video decoder ensures temporal consistency. Our uncertainty-preserving first stage allows us to capture aleatoric uncertainty through ensembling, which is particularly beneficial for extreme weather events with high decoding variability. We achieve state-of-the-art performance in precipitation nowcasting with a compact latent-space rectified flow transformer on the *SEVIR* benchmark and show further performance gains by scaling model size. With *FREUD* and the latent rectified flow model, we aim to push the boundaries of data-driven weather nowcasting.

1. Introduction

Accurate weather forecasting underpins many industries, including energy production [87, 94, 101], agriculture [17, 71, 76], and aviation [22, 65, 72] and is safety-critical during extreme weather events. Nowcasting, the task of producing short-term, high-resolution forecasts over 30 minutes to 12 hours [23, 25, 91], requires rapid updates that traditional numerical weather prediction models cannot provide as their atmospheric simulations are often too slow [1, 84, 91]. Data-driven machine learning (ML) models overcome this speed bottleneck [1, 91], but deterministic formulations often blur predictions by averaging over mul-

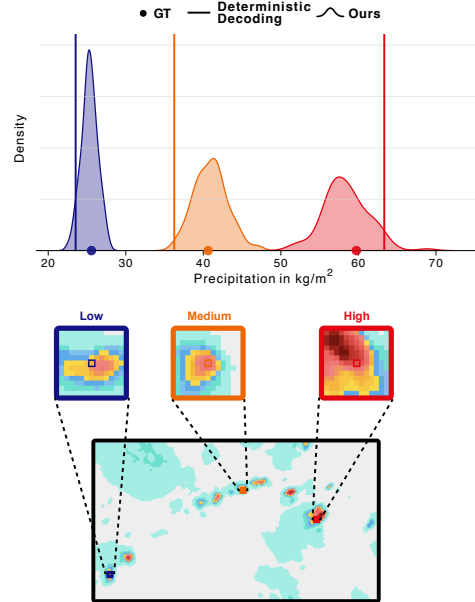


Figure 1. **Reconstruction ensemble distributions** from our generative compression stage for locations with medium (blue), heavy (orange), and extreme (red) precipitation. In light-rain regions, the ensemble shows low variance, while under intense and chaotic rainfall, the spread increases and reliably covers the ground-truth values (dots), whereas the deterministic CasCast decoder [32] (lines) cannot quantify uncertainty and fails to cover ground truth.

iple plausible futures [23, 25, 91]. Sampling from probabilistic generative models instead yields sharp, diverse forecasts whose spread naturally quantifies predictive uncertainty [11, 29, 52, 73, 82], a key requirement in safety-critical settings where the range of plausible outcomes matters as much as the most likely one. Early generative approaches use Generative Adversarial Networks (GANs) [33, 58, 73, 109], but their training instability makes employment difficult and the issue of *mode collapse* [49] limits coverage of rare but crucial weather modes. Hence, effective nowcasting requires *data-driven methods* that are *accurate*, *probabilistic*, *fast*, and *scalable*.

Diffusion and flow-based models [3, 4, 40, 57, 59, 74, 85, 86] overcome GAN-related training instabilities and mode

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collapse, but their iterative sampling in high-dimensional data space is computationally expensive. To reduce this cost, image and video models commonly adopt a two-stage setup: an autoencoder compresses data into a latent space, abstracting away imperceptible details, allowing the generative model to operate on a lower-dimensional representation [24, 74, 90]. While effective for natural images and videos, the current *two-stage design is ill-posed for weather forecasting*. Compression is inherently lossy, and errors that are imperceptible in images may correspond to substantial shifts in predicted precipitation, undermining reliability in safety-critical settings. Moreover, training such autoencoders requires carefully balancing regularization [48], perceptual [108], and adversarial losses [33, 74], where adversarial components can reintroduce instability and degrade coverage of extreme events. Despite these drawbacks, recent nowcasting models [29, 32, 52] continue to rely on this architecture to manage the high dimensionality of weather data, thereby inheriting its limitations.

In this work, we introduce a novel and simple first stage design, tailored to weather nowcasting. Replacing the standard decoder with a rectified flow decoder gives four main benefits: 1) the probabilistic rectified flow framework ensures that predictions are sharp and realistic, without adversarial or perceptual losses, which makes it 2) easy to train, 3) scalable, and 4) allows us to quantify uncertainty in the decoding process. To ensure a smooth and well-behaved latent space, we introduce a novel stochastic tanh regularization, which restricts latents to the range $[-1, 1]$. Unlike KL regularization [48], it requires no loss weighting or architectural changes [48, 74]. We further propose an asymmetric architecture for our first stage, tailored for forecasting. In line with recent image synthesis works [35, 112], we utilize transformers for the first-stage encoder and decoder, thereby avoiding the legacy, inefficient convolutional design [112]. For latent space forecasting, we must avoid *information leakage from future to past* frames in the autoencoder, making sequence-level encoders [55, 89, 95, 103, 110] unsuitable. Further, to support incremental updates without re-encoding an entire sequence, we want frame-wise encoding [29, 32, 54]. However, pure frame-wise first stages cause flickering and *temporal inconsistency* [102].

We therefore propose *FREUD*, a **F**Ramewise **E**ncoder and **U**nited **D**ecoder architecture where each frame is encoded independently by a lightweight transformer, while a hierarchical rectified flow transformer [18, 61, 68] decodes them jointly. FREUD provides a fast, scalable first stage for efficient latent-space generation with temporally consistent reconstructions, without adversarial training or complex loss weighting. Prior diffusion-based nowcasting models either use a *fixed conditioning window* [29, 32, 52], which is vulnerable to missing or corrupted frames [2, 19, 47], or generate future frames auto-regressively [46, 70], which ac-

cumulates errors over time. Robust nowcasting therefore requires flexible, adaptive conditioning.

We achieve this through masking-based diffusion forcing [12, 42], which allows inference with any number of conditioning frames and maintains strong skill even with only two past frames. Recent methods cascade a deterministic forecast with a generative model [32, 67, 105], improving localization but yielding overconfident predictions, whereas we condition solely on observed precipitation for better calibration. Combining this insight with our novel FREUD compression stage and masking-based rectified flow training yields state-of-the-art precipitation nowcasting on SEVIR [93]. Our compact model outperforms prior work in Continuous Ranked Probability Score (CRPS) and perceptual metrics while achieving competitive localization without biasing the generation. We can formulate our contributions as follows:

- **A simple, scalable first stage tailored for weather nowcasting.** We introduce a rectified-flow encoder-decoder with novel stochastic tanh-regularization that achieves higher reconstruction quality without perceptual or adversarial losses, ensuring a smooth, bounded latent space.
- **Frame-wise encoder and united video decoder.** We propose an asymmetric design that encodes frames independently to prevent information leakage and support variable-length inputs, while jointly decoding all frames simultaneously for temporal consistency.
- **Decoding uncertainty quantification.** Our probabilistic decoder design enables multiple reconstructions from the same latent, providing natural estimates of decoding uncertainty, crucial for safety-critical applications.
- **Scalable and robust latent-space nowcasting.** We achieve state-of-the-art precipitation forecasting through masking-based rectified-flow training that handles variable-length conditioning and frame drops, while scaling with model size, ensemble count, and function evaluations to trade latency for accuracy.

2. Related Works

Precipitation Nowcasting Classical Numerical Weather Prediction (NWP) is computationally expensive, making short-term nowcasts difficult [14, 52, 73, 80, 81, 84]. Optical-flow extrapolation methods [16, 31, 75, 99, 100] are faster but cannot capture the formation of new weather patterns [45, 80]. These issues motivate deep-learning approaches, categorized as deterministic or probabilistic. Deterministic models learn spatio-temporal mappings from past to future precipitation [1, 27, 28, 34, 80, 97]. However, they often yield blurry forecasts due to regression losses and also lack uncertainty quantification [29, 52, 64, 73]. Probabilistic approaches use GANs [58, 73, 109] or diffusion models (DMs) [29, 32, 52, 56, 64, 67] to generate weather predictions, producing sharp samples and estimating un-

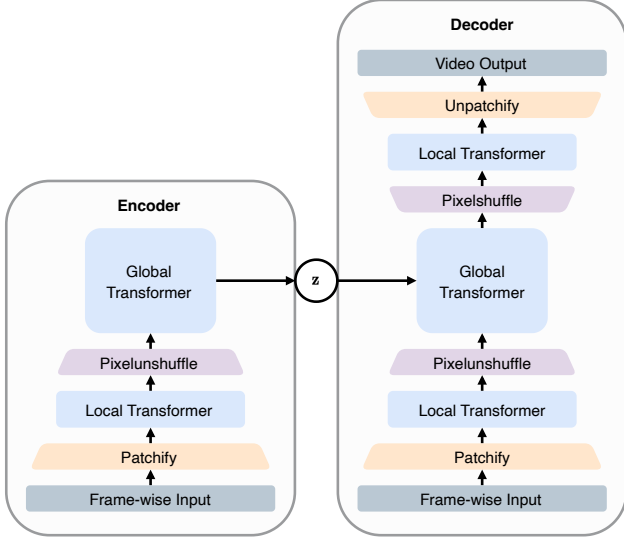


Figure 2. **FREUD architecture.** *Left:* Frame-wise encoder. *Right:* Generative decoder conditioned on encoder latents $\mathbf{z}$.

certainty via Monte-Carlo (MC) sampling. Mode collapse and training instabilities limit GANs [49], making diffusion models the state of the art [29, 32, 52, 105]. Many methods operate in the latent space of a compression autoencoder [29, 32, 52], but ignore compression uncertainty [56]. Moreover, deterministic conditioning [32, 66, 105], often biases the generation [67]. We address these issues with a first stage that quantifies compression uncertainty and enables purely data-driven nowcasting. Further discussion of related works is provided in the Appendix.

Uncertainty Estimation in ML differentiates uncertainty from underdeterminedness of the problem (aleatoric uncertainty) or underspecification of the model (epistemic uncertainty) [44]. Aleatoric uncertainty is important in weather nowcasting due to non-linearities in the atmospheric system [41, 52, 64, 78]. Traditionally, Bayesian Neural Networks [62, 63], MC Dropout [26, 88], and model ensembling [50] are used to estimate uncertainty but are either impractical for large generative models or degrade performance [92]. DMs allow zero-shot aleatoric uncertainty quantification [11, 82] by sampling multiple times given the same conditioning. The variance of this ensemble serves as an estimate of aleatoric uncertainty [11, 29, 52, 64].

Diffusion Autoencoders (DiffAEs) replace the deterministic decoder with a generative model conditioned on encoder outputs [69]. Previous work uses discretized [6, 7, 77, 107] as well as continuous [8, 15, 43, 111] latent encodings to enable downstream generation in the latent space [7, 8, 15, 43, 77, 111]. Compared to standard VAEs, DiffAEs yield better detail reconstruction for images and

more stable training [15, 111]. Earlier video generation methods use frame-wise compression [5, 9, 30, 38, 60, 83, 96], causing temporal inconsistencies [102, 110], or 3D encoder-decoder models [55, 89, 90, 95, 106], with limited capacity [110]. Concurrent DiffAE-based video compressors [103, 110] achieve superior reconstructions over VAE baselines with few sampling steps and strong generative downstream performance. We improve upon these approaches by using a hierarchical Transformer-based architecture, unlocking scalable performance, and introducing a novel regularization. Moreover, our design better suits forecasting by avoiding perceptual losses, encoding frames independently, and assessing reconstruction uncertainty.

3. Method

In the following, we consider sequences of precipitation maps $\mathbf{x}^t \in \mathbb{R}^{C \times H \times W}$, where C , H , and W denote channels, height, and width, respectively. The superscript t indexes time frames, and the subscript i denotes diffusion steps $\mathbf{x}_i$. Given the L^{in} previous precipitation maps, the task is to predict the next L^{out} frames. As we cannot model the chaotic nature of precipitation deterministically, we sample from the conditional distribution $p(\mathbf{x}^{\text{out}}|\mathbf{c})$ of possible future precipitation using a generative model. We follow previous work and cast the precipitation nowcasting problem as a probabilistic spatio-temporal prediction task [27, 29, 105]. Hence, future frames $\mathbf{x}^{\text{out}}$ are sequences of L^{out} precipitation maps while the conditioning $\mathbf{c}$ are L^{in} previous frames.

3.1. Preliminary

Rectified Flow Generative models that transform noise into data have recently gained significant attention, particularly diffusion- [40, 74, 85, 86] and flow-based models [3, 4, 13, 21, 57, 59]. *Rectified Flows* (RF) [24, 59] or *Stochastic Interpolants* [3, 4, 61] generalize these frameworks by mapping samples $\mathbf{x}_0 \sim p_0$ from a simple prior (typically $\mathcal{N}(0, \mathbf{I})$) to data samples $\mathbf{x}_1 \sim p_1$ through the continuous interpolation $\mathbf{x}_i = \alpha_i \mathbf{x}_1 + \sigma_i \mathbf{x}_0$, where α_i increases and σ_i decreases over time, controlling the transition from noise to data. The velocity field is defined as

$$\mathbf{v}(\mathbf{x}_i, i) = \frac{d\mathbf{x}_i}{di} = \dot{\alpha}_i \mathbf{x}_1 + \dot{\sigma}_i \mathbf{x}_0 \quad (1)$$

where $\dot{\alpha}_i$ and $\dot{\sigma}_i$ are time derivatives of α_i and σ_i . This induces a probability flow ODE whose marginal distribution at time i equals $p_i(\mathbf{x})$. Following prior work [24, 57, 61], we adopt the linear interpolant $\alpha_i = i$, $\sigma_i = 1 - i$. A neural network $\mathbf{v}_\theta(\mathbf{x}_i, i)$ with parameters θ is trained to predict the velocity field by minimizing

$$\mathcal{L}_\mathbf{v}(\theta) = \int \mathbb{E}[\|\mathbf{v}_\theta(\mathbf{x}_i, i) - (\dot{\alpha}_i \mathbf{x}_1 + \dot{\sigma}_i \mathbf{x}_0)\|^2] di. \quad (2)$$

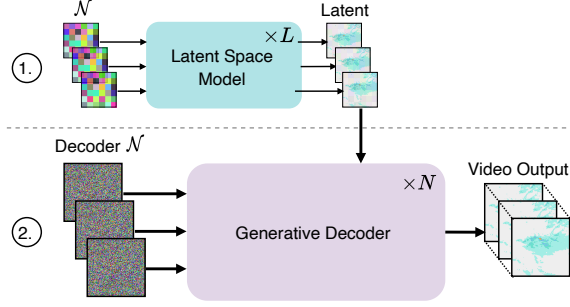


Figure 3. **Inference pipeline.** We first generate weather forecasts in the latent space of our FREUD encoder and then decode the predictions with our generative FREUD decoder.

After training, samples from the prior are transported to the data distribution by numerically integrating the ODE with $\mathbf{v}_\theta(\mathbf{x}_i, i)$ using standard solvers, e.g., Euler integration.

Uncertainty Quantification Following Chan et al. [11], we consider the atmosphere to be a dynamic system $\mathbf{x}^{\text{out}} = \mathcal{F}(\mathbf{c}) + \eta$ with deterministic dynamics $\mathcal{F}(\mathbf{c})$ and irreducible or unmodeled noise η . Even if we can perfectly capture the deterministic dynamics of the system $\mathcal{F}(\mathbf{c})$, there remains an error because of the stochastic component. Therefore, if we can draw Monte-Carlo samples $\mathbf{x}_e^{\text{out}}$ from the distribution, the sample variance

$$\text{Var}(\tilde{\mathbf{x}}^{\text{out}}), \quad \tilde{\mathbf{x}}^{\text{out}} = [\mathbf{x}_e^{\text{out}}]_{e=0}^N, \quad \mathbf{x}_i^{\text{out}} \sim p(\mathbf{x}^{\text{out}}|\mathbf{c}) = \mathcal{F}(\mathbf{c}) + \eta \quad (3)$$

where the subscript $\cdot_e$ indexes independently sampled ensemble members, will converge to the irreducible aleatoric uncertainty $\text{Var}(\eta)$ for $N \rightarrow \infty$ [11]. As we can approximate $p(\mathbf{x}^{\text{out}}|\mathbf{c})$ using a generative model, we can estimate the uncertainty by computing N probabilistic predictions and calculating the ensemble variance [11, 52, 64, 78].

3.2. FREUD

Diffusion- and flow-based generative methods iteratively sample in data space, which is particularly costly for high-resolution weather data. Therefore, following prior work on image and video synthesis, we perform generative modeling in a compressed latent space, which substantially reduces computational complexity [32, 52, 74, 74, 90].

For operational weather nowcasting, a first-stage encoder-decoder should fulfill several key properties. First, **training should be simple and stable**, avoiding adversarial or perceptual objectives. Second, the model must offer **efficient compression** of high-dimensional radar. Third, its latent space should be **smooth and well-structured**, amenable to generative modeling. Fourth, the model should enable **uncertainty estimation** in the decoding process, an essential property for safety-critical applications. Fifth,

frame-wise encoding is desirable to ensure robustness to sensor faults or dropped frames, common in radar networks. Finally, outputs must be **temporally coherent**, since flickering reconstructions across frames would lead to inconsistencies in subsequent forecasts. They motivate our design of the FREUD framework, which features an architecture that provides efficient compression, robustness to sensor faults, and temporally coherent reconstructions.

Architecture Fig. 2 shows our FREUD architecture. To ensure efficiency, we follow recent works [35, 112] and use a fully transformer-based encoder-decoder design.

Encoder. The encoder is implemented as a lightweight transformer that operates frame-wise, allowing each frame to be processed independently. This makes the model robust to missing or corrupted radar frames, which are common in operational weather sensing, and enables online updates as new frames become available without requiring re-encoding of the entire sequence. The frame-wise design further prevents information leakage from future to past frames, which sequence-level encoders would suffer from, thereby preserving the causal structure essential for forecasting.

Decoder. For decoding, we use a transformer-based video decoder that jointly reconstructs all frames to ensure temporal consistency. To cope with the high computational cost of attention in large video tensors, we follow *Hourglass diffusion transformers* [18], and adopt a hierarchical transformer architecture. Here, the spatial resolution is progressively reduced and later restored through pixel-unshuffle and pixel-shuffle operations, which rearrange spatial dimensions into channel dimensions (and vice versa) [79]. We condition the decoder on the encoder latents by concatenating them channel-wise in the bottleneck, guiding the video reconstruction process through the encoded spatial representations. We further improve efficiency through space-time factorized attention [36]. Each block alternates between spatial and temporal attention, followed by a feed-forward network. Spatial attention attends to all tokens within each frame, and temporal attention attends to all tokens in the same spatial positions across all frames.

Finally, we further reduce complexity in high-resolution stages by integrating neighborhood attention [37], which restricts self-attention to local spatial neighborhoods rather than full global context.

Training Paradigm Previous works train autoencoders with a small KL regularization to encourage a smooth latent space, and additional perceptual and adversarial losses to boost sharpness [32, 52, 74]. However, these objectives require careful loss balancing [15, 103, 110, 111], and adversarial training often leads to training instability and mode collapse, while both perceptual and adversarial losses can suppress subtle yet safety-critical details [49, 56]. More-

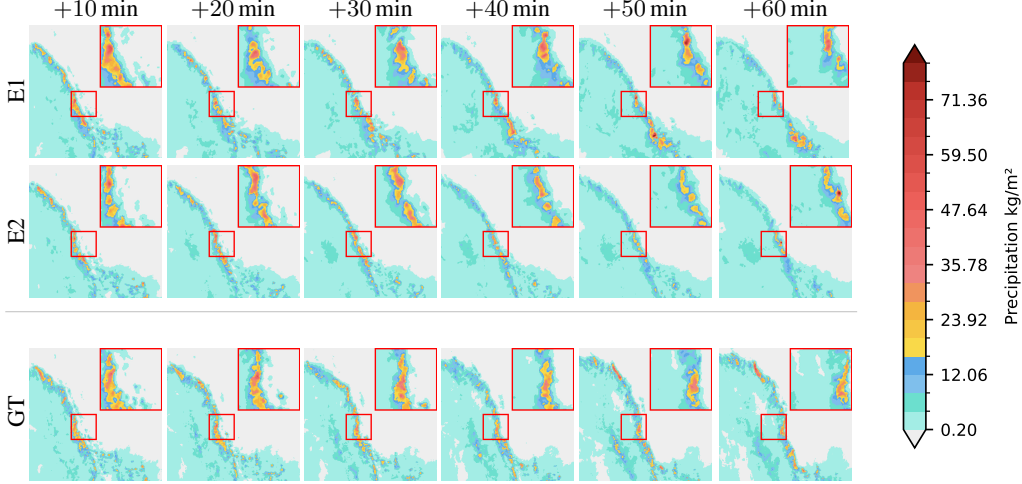


Figure 4. **Forecast ensemble** from our L model. Red rectangles show zoom-ins to highlight differences. Best viewed zoomed in.

over, their decoders are deterministic at inference, providing no estimate of decoding uncertainty. We address these limitations by training our first stage with the rectified flow (RF) loss from Eq. (2). Optimization with this loss is stable and makes perceptual and adversarial objectives obsolete, greatly simplifying training. It further allows us to sample from our decoder during inference, which naturally provides an empirical way to quantify decoding uncertainty through multiple reconstructions. We condition the decoder on the frame-wise latents and train both the encoder and decoder jointly with the RF loss from Eq. (2).

Latent Space Regularization A well-structured latent space is essential for latent space generative modeling [35, 104, 112]. Current autoencoders typically include a small KL regularization term to encourage a smooth latent space [74]. In contrast, our framework does not inherently constrain the latent space, leading to high variance and low density. Thus, in addition to the *unregularized (unreg.)* case, we introduce and ablate two additional regularization schemes to ensure robust latent representations.

KL-regularization (KL-reg.), as used in latent diffusion autoencoders [74], penalizes the divergence between the latent distribution and a standard normal. This encourages smoother and more robust latent representations but introduces a trade-off: stronger KL weights improve regularization at the cost of reconstruction fidelity, requiring loss balancing. Moreover, it requires architectural modifications, as the encoder must predict both the mean and the standard deviation per latent dimension.

Stochastic tanh regularization (T-reg.). We propose *T-reg.* as a simple yet effective alternative to KL-based regularization that achieves all desired properties of a well-behaved latent space: bounded, smooth, and amenable to generation. We constrain the latent values to $[-1, 1]$ via a tanh activa-

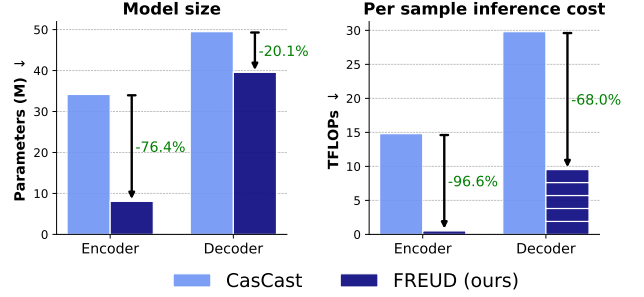


Figure 5. **FREUD efficiency.** Our transformer-based autoencoder has fewer parameters and uses fewer FLOPs for encoding and decoding (5 NFE). This allows for faster training and inference.

tion and add a small Gaussian perturbation $\epsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$ with variance σ^2 to the latent representation, so that

$$\tilde{\mathbf{z}}^t = \tanh(\text{Enc}_\theta(\mathbf{x}^t)) + \sigma \epsilon \quad (4)$$

where Enc_θ denotes the frame-wise FREUD encoder and $\mathbf{x}^t$ the input frame. This stochastic perturbation makes nearby latents decode into similar pixel-space videos, encouraging smoothness and robustness to small changes in the latent space. Unlike KL regularization, *T-reg.* acts purely as an architectural constraint rather than an additional loss, eliminating the need for loss balancing. The final loss for training the FREUD first stage with *T-reg.* is thus just the flow loss from Eq. (2), simplified with the linear schedule to

$$\mathcal{L} = \|\mathbf{v}_\theta(\mathbf{x}_i, i) - (\mathbf{x}_1 - \mathbf{x}_0)\|^2 \quad (5)$$

where $\mathbf{x}_1$ is a data sample and $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$. Similar to [90], we find that a simple outlier punishment is helpful in early training (for details see Appendix).

3.3. Latent Space Nowcasting

With FREUD as our first stage, we perform nowcasting directly in its latent space. Its frame-wise encoder tolerates missing frames, so our latent model must likewise support variable-length conditioning and remain robust under limited past observations. For this we adopt a masking-based training strategy, which teaches the model to infer future dynamics from arbitrary subsets of past frames and allows flexible conditioning lengths during inference, ensuring resilience to missing or corrupted frames.

Training We follow the simple training paradigm of RaMVid [42], where we randomly mask part of each clip and learn to denoise the remaining part. More formally, we split a video of length $T = L^{\text{in}} + L^{\text{out}}$ into conditioning frames $\mathbf{x}_1^{\mathcal{C}}$ and generation frames $\mathbf{x}_1^{\mathcal{G}}$ where $\mathcal{C} \cup \mathcal{G} = \{0, \dots, T-1\}$ and $\mathcal{C} \cap \mathcal{G} = \emptyset$. The final video is given by $\mathbf{x}_1 = \mathbf{x}_1^{\mathcal{C}} \oplus \mathbf{x}_1^{\mathcal{G}}$ where $\oplus$ is a frame selection operation. For every sample we draw k conditioning indices uniformly from $\{1, \dots, K\}$, where the hyperparameter $K < T$ defines the maximum number of conditioning frames. Only the remaining frames with indices in $\mathcal{G} = \{0, \dots, T-1\} \setminus \mathcal{C}$, are noised, so the model input is $\mathbf{x}_i = \mathbf{x}_1^{\mathcal{C}} \oplus \mathbf{x}_i^{\mathcal{G}}$. Randomising the amount of conditioning $|\mathcal{C}| = k$ during training teaches the model to leverage any variable subsets of information to make its predictions. We compute the loss from Eq. (2) only on noisy frames $\mathbf{x}_i^{\mathcal{G}}$. Similar to [42] we define probability $p_{\mathcal{U}}$ with which we train fully unconditional, i.e., set $\mathcal{C} = \emptyset$. This enables classifier-free guidance [39].

Inference At inference we encode the latest L^{in} available observations with the frame-wise encoder and append L^{out} Gaussian latent frames $\mathbf{x}_0^t \in \mathbb{R}^{C_l \times H_l \times W_l}$. This yields a sequence of $T = L^{\text{in}} + L^{\text{out}}$ latent frames, consisting of conditioning latents and noisy targets. The latent rectified flow model then denoises this sequence, and the FREUD decoder maps the denoised latents back to pixel space, producing a forecast for the L^{out} future frames. See Fig. 3 for an overview of the full pipeline. To estimate predictive uncertainty, we form latent-space ensembles by resampling the L^{out} noise latents while keeping the conditioning fixed. Additionally, we can generate decoder ensembles by rerunning the decoding process with different noise initializations for the same latent sequence.

4. Experiments

Following prior work [27, 32], we evaluate our method on the SEVIR benchmark [93], which consists of 20,393 (extreme) weather events, collected from 2017 to 2019. Each event covers a 384×384 km region for a 4 h timespan. Vertically Integrated Liquid (VIL) from the NEXRAD radar mosaic indicates precipitation and is recorded with 1 km spa-

Table 1. **Reconstruction performance** of the framewise CasCast autoencoder, a framewise Diffusion Autoencoder, and FREUD variants with different regularization.

Model	Ens.	RMSE ↓	MAE ↓	SSIM ↑	PSNR ↑	dMAE ↓
CasCast [32]	–	0.022	0.008	0.976	39.153	0.012
T-reg.	1	0.027	0.011	0.970	37.415	0.015
DiffAE	10	0.025	0.010	0.974	38.13	0.012
unreg.	1	0.027	0.009	0.989	37.550	0.013
FREUD	10	0.023	0.008	0.987	38.915	0.012
KL-reg.	1	0.025	0.009	0.992	37.927	0.013
FREUD	10	0.022	0.008	0.987	39.029	0.011
T-reg.	1	0.019	0.008	0.998	40.224	0.011
FREUD	10	0.018	0.007	0.999	41.085	0.010

tial and 5 min temporal resolution. We provide more details on the SEVIR dataset, the computational resources used for experiments, as well as additional experiments on the MeteoNet benchmark [51] in the Appendix.

4.1. FREUD First Stage

First, we evaluate our frame-wise encoder and joint decoder architecture. We follow prior work [15, 103, 110, 111] and report RMSE, MAE, SSIM, and PSNR. We capture temporal smoothness with the MAE of discrete time derivatives (dMAE), and overall predictive uncertainty by the variance (Var) over ensemble members, as defined in Sec. 3.1.

Reconstruction Performance We report reconstruction metrics for FREUD and the prior state-of-the-art frame-wise autoencoder CasCast [32] in Tab. 1. Our *T-reg.* FREUD achieves superior results across all metrics, with a significant gain in SSIM, indicating perceptually accurate reconstructions without adversarial or perceptual losses. It further needs fewer parameters and fewer Floating Point Operations (FLOPs) than the CasCast autoencoder, as shown in Fig. 5. Encoding is 96%, while decoding is 68% faster with five NFE. This efficiency pays off twice: fast encoding accelerates latent-space model training, while efficient encoding and decoding enable rapid forecast updates during inference, which is crucial for operational nowcasting. In summary, FREUD is not only easier to train, but also more compute-efficient, and delivers higher-fidelity reconstructions, even with only one ensemble member.

Decoding Uncertainty Through the generative decoder, FREUD can sample multiple realizations given the same conditioning signal. As described in Sec. 3.1, we estimate decoding uncertainty from the variance across reconstructions. Fig. 7 shows a five-member ensemble of FREUD reconstructions for both normal and extreme weather events along with corresponding ensemble variance maps. While

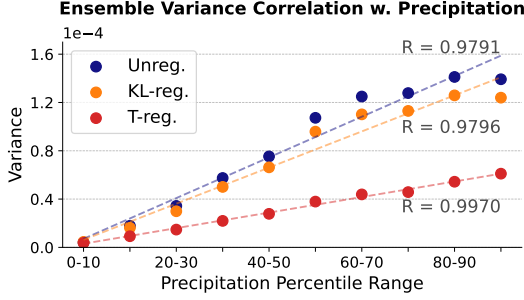


Figure 6. **Correlation between precipitation intensity and decoding variance.** All FREUD variants show a strong linear relationship, with *T-reg.* achieving the highest correlation.

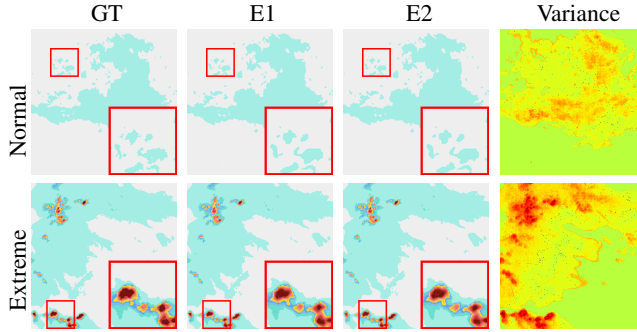


Figure 7. **Qualitative reconstructions** of a normal and an extreme weather event, with corresponding ensemble members and ensemble variance. The red rectangle highlights a zoomed-in area for visualizing differences in details. Best viewed zoomed in.

reconstructions are highly similar overall, they differ in small-scale structures, particularly in regions of intense precipitation. Unlike natural image synthesis, where such variations are imperceptible, in weather forecasting they reflect genuine differences in high-impact regions. The variance maps reveal a clear correlation between precipitation intensity and reconstruction variance, consistent with the non-linearity and chaotic dynamics of severe weather. This shows that ensemble-based reconstructions capture meaningful variability precisely in regions of heightened uncertainty, providing valuable insight for safety-critical extreme events. Fig. 6 further confirms a strong linear relationship between precipitation percentile and intra-ensemble variance across all FREUD variants, with the *T-reg.* model showing the highest correlation ($r = 0.997$). In summary, these findings confirm that first-stage ensembling effectively identifies dynamically complex weather, providing meaningful and spatially localized uncertainty estimates. Additional experiments inserting abnormal precipitation patterns, similar to [11], as well as more qualitative results are presented in the Appendix.

Table 2. **SEVIR benchmark** comparison of our method with various deterministic and probabilistic models. Baselines are sourced from CasCast [32]. For † the model is trained on a downsampled dataset with a size of 128 since they found that training on the original dataset does not yield high-quality predictions.

Method	CRPS↓	SSIM↑	HSS↑	CSI↑
ConvLSTM [80] (NeurIPS '15)	0.0264	0.7749	0.5232	0.4102
PredRNN [97] (TPAMI '22)	0.0271	0.7497	0.5192	0.4045
PhyDNet [34] (CVPR '20)	0.0253	0.7649	0.5311	0.4198
SimVP [28] (CVPR '22)	0.0259	0.7772	0.5280	0.4153
EarthFormer [27] (NeurIPS '22)	0.0251	0.7756	0.5411	0.4310
NowcastNet [109] (Nature '23)	0.0283	0.5696	0.5365	0.4152
PreDiff† [29] (NeurIPS '23)	0.0202	0.7648	0.4914	0.3875
CasCast [32] (ICML '24)	0.0202	0.7797	0.5602	0.4401
FREUD + LSM-L (ours)	0.0190	0.7841	0.5011	0.3864
↳ with cfg	0.0192	0.7937	0.5537	0.4277

4.2. Forecasting performance

Given the FREUD first stage, we train a latent space model for forecasting. We follow prior works [27, 32] and use $L^{\text{in}} = 13$ frames (65 min) of previous precipitation to predict the next $L^{\text{out}} = 12$ frames (60 min). Unless specified otherwise, we use ten forecasting ensemble members similar to prior work [29, 32, 52, 105]. We use the evaluation pipeline provided by Gong et al. [32] and evaluate probabilistic nowcasting performance with the Continuous Ranked Probability Score (CRPS), perceptual realism with the Structural Similarity Index Measure (SSIM), and report Heidke Skill Score (HSS) and Critical Success Index (CSI). HSS expresses improvement over random chance, with 1 indicating a perfect forecast and 0 indicating no improvement over guessing. CSI quantifies the hit rate while ignoring true negatives, making it robust when no events dominate. Both scores are averaged across six precipitation/no precipitation thresholds. We provide more details in the Appendix.

Comparison to State-of-the-Art Tab. 2 quantitatively compares our method to current state-of-the-art weather nowcasting methods on the SEVIR benchmark. Our Large Latent Space Model (LSM-L) combined with the *T-reg.* FREUD first stage achieves state-of-the-art CRPS and SSIM, outperforming all existing methods in both distribution coverage and perceptual quality. Our method provides significant gains, given that improvements on SEVIR are typically small. For instance, CasCast [32] improves over PreDiff [29] by 0% CRPS and +1.95% SSIM, whereas our method improves over CasCast by +5.94% CRPS and +1.80% SSIM. CasCast reports its state-of-the-art results using Classifier-Free Guidance (CFG) [39], which also improves localization metrics in our model. With CFG, our method becomes competitive on HSS and CSI, being only outperformed by CasCast on HSS. However, we observe

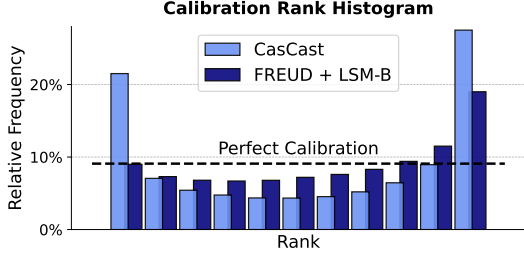


Figure 8. **Calibration Rank Histogram** for our pipeline. Our rank histogram is flatter, indicating improved calibration compared to CasCast [32].

that CFG systematically increases predicted precipitation for both CasCast and our method, suggesting that CFG-based improvements in localization may stem from this simple shift rather than better modeling. We therefore consider CFG to be flawed for nowcasting and further ablate its effects in the Appendix. Qualitative results in Fig. 4 show that our predictions are visually consistent with the ground truth and capture realistic variations in localized precipitation patterns. We provide more samples in the Appendix.

Calibration Similar to previous work [10, 52, 53], we use ensemble rank histograms to assess calibration: a uniform shape indicates well-calibrated uncertainty, whereas a U-shape reflects overconfidence. As shown in Fig. 8, our ensemble is substantially closer to uniform, indicating better calibration. In contrast, CasCasts rank histogram indicates strong overconfidence. Quantitative assessment via the reliability index (RI) [20, 98], with 1,000 bootstrap resamples to compute a 95% confidence interval, confirms our approach achieves a significantly better reliability error ($RI = 0.135 \pm 0.001$) than CasCast ($RI = 0.312 \pm 0.001$).

Scalability We show the scalability of our approach by training models of different sizes, with quantitative results reported in Tab. 3. Similar to trends observed in image synthesis, we find that larger latent-space models consistently improve performance across all metrics. Notably, even our smallest model (LSM-S) achieves competitive performance despite using only a fraction of the parameters of prior work. Our largest model (LSM-L) achieves the best overall results, confirming that our approach scales favorably with model capacity.

Deterministic Prior We achieve top-3 CSI and 2nd-best HSS without deterministic priors. Intuitively, conditioning on such priors reduces diversity by lowering conditional variance, which can improve localization of mean predictions if correct but collapses the distribution to incorrect modes otherwise. Our RF training enables zero-shot deterministic conditioning by initializing generation via noisy

Table 3. **Scaling Analysis** of the forecasting latent space flow model. Localization metrics and CRPS consistently improve with larger model sizes. All models outperform CasCast for CRPS.

Model	Num. Params	CRPS ↓	SSIM ↑	HSS ↑	CSI ↑
CasCast [32]	309M	0.0202	0.7797	0.5602	0.4401
Ours LSM-S	44M	0.0200	0.7742	0.4514	0.3485
↳ with cfg		0.0201	0.7887	0.5155	0.3962
Ours LSM-B	141M	0.0196	0.7828	0.4923	0.3805
↳ with cfg		0.0204	0.7987	0.5256	0.4041
Ours LSM-L	473M	0.0190	0.7841	0.5011	0.3864
↳ with cfg		0.0192	0.7937	0.5537	0.4277

Model	det. prior	i	CRPS ↓	HSS ↑	CSI-M ↑
Earthformer	na	na	0.0251	0.5411	0.4310
CasCast	✓	✗	0.0202	0.5602	0.4401
Ours	✗	0	0.0190	0.5011	0.3864
	✓	0.2	0.0198	0.5714	0.4444
	✓	0.5	0.0219	0.5735	0.4455
	✓	0.75	0.0233	0.5598	0.4338

Table 4. **Training-free deterministic conditioning.** Localization performance on SEVIR improves when introducing deterministic priors while distribution coverage deteriorates. $i = 1$ corresponds to a fully deterministic and $i = 0$ to a fully generative prediction.

Table 5. Comparison of B-LSMs in our regularized latent spaces. *T-reg.* outperforms *unreg.* and *KL-reg.* in terms of CRPS and SSIM but performs worse for localization metrics.

Regularization	CRPS ↓	SSIM ↑	HSS ↑	CSI ↑
Unreg.	0.0222	0.7630	0.5139	0.3956
KL-reg.	0.0201	0.7790	0.4993	0.3892
T-reg.	0.0196	0.7828	0.4923	0.3805

Earthformer [27] predictions, where the noise level allows users to express confidence in the deterministic prediction. Tab. 4 confirms the described trade-off: relying on deterministic predictions allows better localization but worse coverage and calibration.

4.3. Ablations

Regularization As shown in Tab. 1, *T-reg.* FREUD delivers the best reconstruction quality and the most meaningful uncertainty estimates (Fig. 6). *T-reg.* also produces a more compact and structured latent space (see Appendix), whereas other variants exhibit higher variance and lower density. Tab. 5 shows that this improved structure translates into better downstream performance: B-LSMs trained on *T-reg.* latents achieve superior CRPS and SSIM compared to those trained on other latent spaces.

Joint Video Decoding We ablate the effect of united (joint) decoding in Tab. 1. While a frame-wise DiffAE is competitive with the frame-wise CasCast autoencoder [32], it performs noticeably worse than FREUD. Using the same

regularization settings, FREUD reduces dMAE by 33%, indicating improved temporal consistency. Qualitative video samples in the supplementary material further confirm this effect: frame-wise decoding exhibits small fluctuations and flickering artifacts, whereas united decoding produces smoother and more temporally stable reconstructions.

5. Conclusion

We introduce FREUD, a simple and scalable first-stage compression framework. FREUD is built on a frame-wise transformer encoder and a probabilistic rectified-flow video decoder. This design eliminates the need for perceptual or adversarial losses, produces sharper and more stable reconstructions, and, most importantly, allows for direct uncertainty estimation at the decoding stage. When combined with a latent-space rectified-flow transformer, our pipeline achieves state-of-the-art precipitation nowcasting on SE-VIR. It offers strong calibration, flexible conditioning, and favorable scaling with model size, ensemble size, and test-time computation. By capturing first- and second-stage uncertainties while remaining purely data-driven, our approach paves the way for reliable short-term precipitation forecasting, without relying on deterministic physics-based predictors. We discuss remaining limitations and potential societal impacts in the Appendix.

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References

- [1] Shreya Agrawal, Luke Barrington, Carla Bromberg, John Burge, Cenk Gazen, and Jason Hickey. Machine learning for precipitation nowcasting from radar images. *arXiv preprint arXiv:1912.12132*, 2019. 1, 2
- [2] Ali M. Al-Saegh, Taha A. Elwi, Osama A. Abdullah, Aduwati Sali, and Abdulmajeed H. J. Aljumaily. Rainfall Effect on Satellite Communications in Mosul at Frequencies above 10 GHz. In *2021 7th International Conference on Space Science and Communication (IconSpace)*, pages 318–322, 2021. 2
- [3] Michael Samuel Albergo and Eric Vanden-Eijnden. Building normalizing flows with stochastic interpolants. In *ICLR*, 2023. 1, 3
- [4] Michael S Albergo, Nicholas M Boffi, and Eric Vanden-Eijnden. Stochastic interpolants: A unifying framework for flows and diffusions. *arXiv preprint arXiv:2303.08797*, 2023. 1, 3
- [5] Jie An, Songyang Zhang, Harry Yang, Sonal Gupta, Jia-Bin Huang, Jiebo Luo, and Xi Yin. Latent-Shift: Latent Diffusion with Temporal Shift for Efficient Text-to-Video Generation, 2023. 3
- [6] Roman Bachmann, Oğuzhan Fatih Kar, David Mizrahi, Ali Garjani, Mingfei Gao, David Griffiths, Jiaming Hu, Afshin Dehghan, and Amir Zamir. 4M-21: An Any-to-Any Vision Model for Tens of Tasks and Modalities. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. 3
- [7] Roman Bachmann, Jesse Allardice, David Mizrahi, Enrico Fini, Oğuzhan Fatih Kar, Elmira Amirloo, Alaaeldin El-Nouby, Amir Zamir, and Afshin Dehghan. FlexTok: Resampling Images into 1D Token Sequences of Flexible Length, 2025. 3
- [8] James Betker, Gabriel Goh, Li Jing, † TimBrooks, Jianfeng Wang, Linjie Li, † LongOuyang, † JuntangZhuang, † JoyceLee, † YufeiGuo, † WesamManassra, † PrafullaDhariwal, † CaseyChu, † YunxinJiao, and Aditya Ramesh. Improving Image Generation with Better Captions. In *Computer Science*, 2023. 3
- [9] Andreas Blattmann, Robin Rombach, Huan Ling, Tim Dockhorn, Seung Wook Kim, Sanja Fidler, and Karsten Kreis. Align Your Latents: High-Resolution Video Synthesis with Latent Diffusion Models. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 22563–22575, Vancouver, BC, Canada, 2023. IEEE. 3
- [10] Rüdiger Brecht and Alex Bihlo. Towards replacing precipitation ensemble predictions systems using machine learning. *Atmospheric Science Letters*, n/a(n/a):e1262, 2024. 8
- [11] Matthew A. Chan, Maria J. Molina, and Christopher A. Metzler. Hyper-Diffusion: Estimating Epistemic and Aleatoric Uncertainty with a Single Model, 2024. 1, 3, 4, 7
- [12] Boyuan Chen, Diego Marti Monso, Yilun Du, Max Simchowitz, Russ Tedrake, and Vincent Sitzmann. Diffusion Forcing: Next-token prediction meets full-sequence diffusion, 2024. 2
- [13] Ricky TQ Chen, Yulia Rubanova, Jesse Bettencourt, and David K Duvenaud. Neural ordinary differential equations. *Advances in neural information processing systems*, 31, 2018. 3
- [14] Shengchao Chen, Guodong Long, Jing Jiang, Dikai Liu, and Chengqi Zhang. Foundation Models for Weather and Climate Data Understanding: A Comprehensive Survey, 2023. 2
- [15] Yinbo Chen, Rohit Girdhar, Xiaolong Wang, Sai Saketh Rambhatla, and Ishan Misra. Diffusion Autoencoders are Scalable Image Tokenizers, 2025. 3, 4, 6
- [16] P. Cheung and H.Y. Yeung. Application of optical-flow technique to significant convection nowcast for terminal areas in Hong Kong. In *The 3rd WMO International Symposium on Nowcasting and Very Short-Range Forecastin*, Rio de Janeiro, Brazil, 2012. WMO. 2
- [17] Brian Collins, Yunru Lai, Uwe Grewer, Steve Attard, Justin Sexton, and Keith G. Pembleton. Evaluating the impact of weather forecasts on productivity and environmental footprint of irrigated maize production systems. *Science of The Total Environment*, 954:176368, 2024. 1
- [18] Katherine Crowson, Stefan Andreas Baumann, Alex Birch, Tanishq Mathew Abraham, Daniel Z. Kaplan, and Enrico Shippole. Scalable High-Resolution Pixel-Space Image Synthesis with Hourglass Diffusion Transformers. In *Proceedings of the 41st International Conference on Machine Learning*, pages 9550–9575. PMLR, 2024. 2, 4
- [19] Saurabh Das, Animesh Maitra, and Ashish K. Shukla. Rain Attenuation Modeling in the 10-100 GHz Frequency using Drop Size Distributions for different Climatic Zones in Tropical India. *Progress In Electromagnetics Research B*, 25:211–224, 2010. 2
- [20] Luca Delle Monache, Joshua P. Hacker, Yongmei Zhou, Xingxiu Deng, and Roland B. Stull. Probabilistic aspects of meteorological and ozone regional ensemble forecasts. *Journal of Geophysical Research: Atmospheres*, 111(D24): 2005JD006917, 2006. 8
- [21] Laurent Dinh, Jascha Sohl-Dickstein, and Samy Bengio. Density estimation using real nvp. In *International Conference on Learning Representations*, 2017. 3
- [22] Gabriele Enea, Tom Reynolds, Mark Weber, Ramon Dalmau Codina, and Dirk Schaefer. Analysis of weather-driven air traffic management challenges for major us and european airports. In *SESAR Innovation Days*, Rome, 2024. SESAR Joint Undertaking. 1
- [23] Lasse Espeholt, Shreya Agrawal, Casper Sønderby, Manoj Kumar, Jonathan Heek, Carla Bromberg, Cenk Gazen, Rob Carver, Marcin Andrychowicz, Jason Hickey, et al. Deep learning for twelve hour precipitation forecasts. *Nature communications*, 13(1):5145, 2022. 1
- [24] Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, et al. Scaling rectified flow transformers for high-resolution image synthesis. In *ICML*, 2024. 2, 3
- [25] Jesús García Fernández and Siamak Mehrkanoon. Broad-unet: Multi-scale feature learning for nowcasting tasks. *Neural Networks*, 144:419–427, 2021. 1

- [26] Yarin Gal and Zoubin Ghahramani. Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning, 2016. [arXiv:1506.02142](#). 3
- [27] Zhihan Gao, Xingjian Shi, Hao Wang, Yi Zhu, Yuyang Bernie Wang, Mu Li, and Dit-Yan Yeung. Earthformer: Exploring space-time transformers for earth system forecasting. *Advances in Neural Information Processing Systems*, 35:25390–25403, 2022. 2, 3, 6, 7, 8
- [28] Zhangyang Gao, Cheng Tan, Lirong Wu, and Stan Z Li. Simvp: Simpler yet better video prediction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3170–3180, 2022. 2, 7
- [29] Zhihan Gao, Xingjian Shi, Boran Han, Hao Wang, Xiaoyong Jin, Danielle Maddix, Yi Zhu, Mu Li, and Yuyang (Bernie) Wang. Prediff: Precipitation nowcasting with latent diffusion models. In *Advances in Neural Information Processing Systems*, pages 78621–78656. Curran Associates, Inc., 2023. 1, 2, 3, 7
- [30] Songwei Ge, Seungjun Nah, Guilin Liu, Tyler Poon, Andrew Tao, Bryan Catanzaro, David Jacobs, Jia-Bin Huang, Ming-Yu Liu, and Yogesh Balaji. Preserve Your Own Correlation: A Noise Prior for Video Diffusion Models. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 22873–22884, Paris, France, 2023. IEEE. 3
- [31] Urs Germann and Isztar Zawadzki. Scale-Dependence of the Predictability of Precipitation from Continental Radar Images. Part I: Description of the Methodology. *Monthly Weather Review*, 130(12):2859–2873, 2002. 2
- [32] Junchao Gong, Lei Bai, Peng Ye, Wanghan Xu, Na Liu, Jianhua Dai, Xiaokang Yang, and Wanli Ouyang. CasCast: Skillful high-resolution precipitation nowcasting via cascaded modelling. In *Proceedings of the 41st International Conference on Machine Learning*, pages 15809–15822, Vienna, Austria, 2024. JMLR.org. 1, 2, 3, 4, 6, 7, 8
- [33] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc., 2014. 1, 2
- [34] Vincent Le Guen and Nicolas Thome. Disentangling physical dynamics from unknown factors for unsupervised video prediction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11474–11484, 2020. 2, 7
- [35] Ming Gui, Johannes Schusterbauer, Timy Phan, Felix Krause, Josh Susskind, Miguel Angel Bautista, and Björn Ommer. Adapting self-supervised representations as a latent space for efficient generation, 2025. 2, 4, 5
- [36] William Harvey, Saeid Naderiparizi, Vaden Masrani, Christian Weilbach, and Frank Wood. Flexible diffusion modeling of long videos. *arXiv preprint arXiv:2205.11495*, 2022. 4
- [37] Ali Hassani, Steven Walton, Jiachen Li, Shen Li, and Humphrey Shi. Neighborhood Attention Transformer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6185–6194, 2023. 4
- [38] Yingqing He, Tianyu Yang, Yong Zhang, Ying Shan, and Qifeng Chen. Latent Video Diffusion Models for High-Fidelity Long Video Generation, 2023. 3
- [39] Jonathan Ho and Tim Salimans. Classifier-Free Diffusion Guidance, 2022. 6, 7
- [40] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33:6840–6851, 2020. 1, 3
- [41] Yang Hong, Kuo-lin Hsu, Hamid Moradkhani, and Soroosh Sorooshian. Uncertainty quantification of satellite precipitation estimation and Monte Carlo assessment of the error propagation into hydrologic response. *Water Resources Research*, 42(8), 2006. 3
- [42] Tobias Höppe, Arash Mehrjou, Stefan Bauer, Didrik Nielsen, and Andrea Dittadi. Diffusion models for video prediction and infilling. *arXiv preprint arXiv:2206.07696*, 2022. 2, 6
- [43] Drew A. Hudson, Daniel Zoran, Mateusz Malinowski, Andrew K. Lampinen, Andrew Jaegle, James L. McClelland, Loic Matthey, Felix Hill, and Alexander Lerchner. SODA: Bottleneck Diffusion Models for Representation Learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23115–23127, 2024. 3
- [44] Eyke Hüllermeier and Willem Waegeman. Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods. *Machine Learning*, 110(3):457–506, 2021. 3
- [45] Junhwa Hur and Stefan Roth. Iterative Residual Refinement for Joint Optical Flow and Occlusion Estimation, 2019. 2
- [46] Wenjun Jiang, Xi Zhong, and Jize Zhang. TC-Diffusion: A diffusion-based probabilistic tropical cyclone model with application to typhoon wind hazard assessment. *Reliability Engineering & System Safety*, 251:110350, 2024. 2
- [47] John D. Kanellopoulos, Stelios G. Koukoulas, Nicolaos J. Kolliopoulos, Christos N. Capsalis, and Spyros G. Vettouras. Rain attenuation problems affecting the performance of microwave communication systems. *Annales des Télécommunications*, 45(7):437–451, 1990. 2
- [48] Diederik P. Kingma and Max Welling. Auto-encoding variational bayes. In *2nd International Conference on Learning Representations, ICLR 2014, Banff, AB, Canada, April 14-16, 2014, Conference Track Proceedings*, 2014. 2
- [49] Youssef Kossale, Mohammed Airaj, and Aziz Darouiichi. Mode Collapse in Generative Adversarial Networks: An Overview. In *2022 8th International Conference on Optimization and Applications (ICOA)*, pages 1–6, 2022. 1, 3, 4
- [50] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Proceedings of the 31st International Conference on Neural Information Processing Systems*, pages 6405–6416, Red Hook, NY, USA, 2017. Curran Associates Inc. 3
- [51] Gwennaëlle Larvor, Léa Berthomier, Vincent Chabot, Brice Le Pape, Bruno Pradel, and Lior Perez. Meteonet, an open reference weather dataset by meteo-france. [URL https://meteonet.umr-cnrm.fr/](https://meteonet.umr-cnrm.fr/), 2020. 6

- [52] Jussi Leinonen, Ulrich Hamann, Daniele Nerini, Urs Germann, and Gabriele Franch. Latent diffusion models for generative precipitation nowcasting with accurate uncertainty quantification. *arXiv preprint arXiv:2304.12891*, 2023. 1, 2, 3, 4, 7, 8
- [53] Lizao Li, Robert Carver, Ignacio Lopez-Gomez, Fei Sha, and John Anderson. Generative emulation of weather forecast ensembles with diffusion models. *Science Advances*, 10(13):eadk4489, 2024. 8
- [54] Weidong Li, Baoxiang Pan, Tiejian Li, Congyi Nai, Zhaoxi Li, Bo Lu, Qingyun Duan, and Ming Pan. Latent Diffusion Model for Quantitative Precipitation Estimation and Forecast at km Scale. *ESS Open Archive*, 2024. 2
- [55] Zongjian Li, Bin Lin, Yang Ye, Liuhan Chen, Xinhua Cheng, Shenghai Yuan, and Li Yuan. WF-VAE: Enhancing Video VAE by Wavelet-Driven Energy Flow for Latent Video Diffusion Model, 2025. 2, 3
- [56] Xudong Ling, Chaorong Li, Fengqing Qin, Peng Yang, and Yuanyuan Huang. SRNDiff: Short-term Rainfall Nowcasting with Condition Diffusion Model, 2024. 2, 3, 4
- [57] Yaron Lipman, Ricky TQ Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *ICLR*, 2023. 1, 3
- [58] Hong-Bin Liu and Ickjai Lee. MPL-GAN: Toward Realistic Meteorological Predictive Learning Using Conditional GAN. *IEEE Access*, 8:93179–93186, 2020. 1, 2
- [59] Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow, 2022. 1, 3
- [60] Zhengxiong Luo, Dayou Chen, Yingya Zhang, Yan Huang, Liang Wang, Yujun Shen, Deli Zhao, Jingren Zhou, and Tieniu Tan. Videofusion: Decomposed diffusion models for high-quality video generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10209–10218, 2023. 3
- [61] Nanye Ma, Mark Goldstein, Michael S. Albergo, Nicholas M. Boffi, Eric Vanden-Eijnden, and Saining Xie. SiT: Exploring Flow and Diffusion-based Generative Models with Scalable Interpolant Transformers, 2024. 2, 3
- [62] David J. C. MacKay. *Bayesian methods for adaptive models*. phd, California Institute of Technology, 1992. 3
- [63] David J. C. MacKay. A Practical Bayesian Framework for Backpropagation Networks. *Neural Computation*, 4(3): 448–472, 1992. 3
- [64] Congyi Nai, Baoxiang Pan, Xi Chen, Qiuhong Tang, Guangheng Ni, Qingyun Duan, Bo Lu, Ziniu Xiao, and Xingcai Liu. Reliable precipitation nowcasting using probabilistic diffusion models. *Environmental Research Letters*, 19(3):034039, 2024. 2, 3, 4
- [65] Juan Nunez-Portillo, Alfonso Valenzuela, Antonio Franco, and Damián Rivas. Predicting air traffic congestion under uncertain adverse weather. *Aerospace*, 11(3):240, 2024. 1
- [66] Jaideep Pathak, Shashank Subramanian, Peter Harrington, Sanjeev Raja, Ashesh Chattopadhyay, Morteza Mardani, Thorsten Kurth, David Hall, Zongyi Li, Kamyar Azizzadenesheli, Pedram Hassanzadeh, Karthik Kashinath, and Animesh Anandkumar. FourCastNet: A Global Data-driven High-resolution Weather Model using Adaptive Fourier Neural Operators, 2022. 3
- [67] Jaideep Pathak, Yair Cohen, Piyush Gard, Peter Harrington, Noah Brenowitz, Dale Durran, Morteza Mardani, Arash Vahdat, Shaoming Xu, Karthik Kashinath, and Mike Pritchard. Kilometer-Scale Convection Allowing Model Emulation using Generative Diffusion Modeling, 2024. 2, 3
- [68] William Peebles and Saining Xie. Scalable Diffusion Models with Transformers, 2023. 2
- [69] Konpat Preechakul, Nattanat Chatthee, Suttisak Widadwongsa, and Supasorn Suwajanakorn. Diffusion Autoencoders: Toward a Meaningful and Decodable Representation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10619–10629, 2022. 3
- [70] Ilan Price, Alvaro Sanchez-Gonzalez, Ferran Alet, Tom R. Andersson, Andrew El-Kadi, Dominic Masters, Timo Ewalds, Jacklynn Stott, Shakir Mohamed, Peter Battaglia, Remi Lam, and Matthew Willson. GenCast: Diffusion-based ensemble forecasting for medium-range weather, 2024. 2
- [71] Francisco Raimundo, André Glória, and Pedro Sebastião. Prediction of weather forecast for smart agriculture supported by machine learning. In *2021 IEEE World AI IoT Congress (AIoT)*, pages 0160–0164, 2021. 1
- [72] Azizul Azhar Ramli, Mohammad Rabiul Islam, Mohd Farhan Md. Fudzee, Mohamad Aizi Salamat, and Shahreen Kasim. A practical weather forecasting for air traffic control system using fuzzy hierarchical technique. In *Recent Advances on Soft Computing and Data Mining*, pages 99–109, Cham, 2014. Springer International Publishing. 1
- [73] Suman Ravuri, Karel Lenc, Matthew Willson, Dmitry Kangin, Remi Lam, Piotr Mirowski, Megan Fitzsimons, Maria Athanassiadou, Sheleem Kashem, Sam Madge, et al. Skilful precipitation nowcasting using deep generative models of radar. *Nature*, 597(7878):672–677, 2021. 1, 2
- [74] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10684–10695, 2022. 1, 2, 3, 4, 5
- [75] Hidetomo Sakaino. Spatio-Temporal Image Pattern Prediction Method Based on a Physical Model With Time-Varying Optical Flow. *IEEE Transactions on Geoscience and Remote Sensing*, 51(5):3023–3036, 2013. 2
- [76] Danna Shen, Xiaofeng Zhao, Leyi Chai, Zhuanzhuan Guo, and Chunxiang Leng. Analysis of the agricultural economic value of a weather forecasting service based on a survey of peasant households in chinese provinces. *Humanities and Social Sciences Communications*, 11(1):1–7, 2024. 1
- [77] Jie Shi, Chenfei Wu, Jian Liang, Xiang Liu, and Nan Duan. DiVAE: Photorealistic Images Synthesis with Denoising Diffusion Decoder, 2022. 3
- [78] Jimeng Shi, Bowen Jin, Jiawei Han, and Giri Narasimhan. CoDiCast: Conditional Diffusion Model for Weather Prediction with Uncertainty Quantification, 2024. 3, 4

- [79] Wenzhe Shi, Jose Caballero, Ferenc Huszar, Johannes Totz, Andrew P. Aitken, Rob Bishop, Daniel Rueckert, and Zehan Wang. Real-Time Single Image and Video Super-Resolution Using an Efficient Sub-Pixel Convolutional Neural Network. In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1874–1883, Las Vegas, NV, USA, 2016. IEEE. 4
- [80] Xingjian Shi, Zhourong Chen, Hao Wang, Dit-Yan Yeung, Wai-Kin Wong, and Wang-chun Woo. Convolutional lstm network: A machine learning approach for precipitation nowcasting. *Advances in neural information processing systems*, 28, 2015. 2, 7
- [81] Xingjian Shi, Zhihan Gao, Leonard Lausen, Hao Wang, Dit-Yan Yeung, Wai-kin Wong, and Wang-chun Woo. Deep learning for precipitation nowcasting: A benchmark and a new model. *Advances in neural information processing systems*, 30, 2017. 2
- [82] Dule Shu and Amir Barati Farimani. Zero-Shot Uncertainty Quantification using Diffusion Probabilistic Models, 2024. 1, 3
- [83] Uriel Singer, Adam Polyak, Thomas Hayes, Xi Yin, Jie An, Songyang Zhang, Qiyuan Hu, Harry Yang, Oron Ashual, Oran Gafni, Devi Parikh, Sonal Gupta, and Yaniv Taigman. Make-A-Video: Text-to-Video Generation without Text-Video Data, 2022. 3
- [84] Casper Kaae Sønderby, Lasse Espeholt, Jonathan Heek, Mostafa Dehghani, Avital Oliver, Tim Salimans, Shreya Agrawal, Jason Hickey, and Nal Kalchbrenner. Metnet: A neural weather model for precipitation forecasting. *arXiv preprint arXiv:2003.12140*, 2020. 1, 2
- [85] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502*, 2020. 1, 3
- [86] Yang Song and Stefano Ermon. Generative modeling by estimating gradients of the data distribution. *Advances in neural information processing systems*, 32, 2019. 1, 3
- [87] Simone Sperati, Stefano Alessandrini, and Luca Delle Monache. An application of the ECMWF Ensemble Prediction System for short-term solar power forecasting. *Solar Energy*, 133:437–450, 2016. 1
- [88] Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. Dropout: A Simple Way to Prevent Neural Networks from Overfitting. *Journal of Machine Learning Research*, 15(56):1929–1958, 2014. 3
- [89] Anni Tang, Tianyu He, Junliang Guo, Xinle Cheng, Li Song, and Jiang Bian. VidTok: A Versatile and Open-Source Video Tokenizer, 2024. 2, 3
- [90] The Movie Gen team @ Meta. Movie Gen: A Cast of Media Foundation Models, 2024. 2, 3, 4, 5
- [91] Kevin Trebing, Tomasz Stanczyk, and Siamak Mehrkanoon. Smaat-unet: Precipitation nowcasting using a small attention-unet architecture. *Pattern Recognition Letters*, 145:178–186, 2021. 1
- [92] Matias Valdenegro-Toro and Daniel Saromo Mori. A Deeper Look into Aleatoric and Epistemic Uncertainty Disentanglement. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 1508–1516, 2022. ISSN: 2160-7516. 3
- [93] Mark Veillette, Siddharth Samsi, and Chris Mattioli. SE-VIR : A Storm Event Imagery Dataset for Deep Learning Applications in Radar and Satellite Meteorology. In *Advances in Neural Information Processing Systems*, pages 22009–22019. Curran Associates, Inc., 2020. 2, 6
- [94] Huai-zhi Wang, Gang-qiang Li, Gui-bin Wang, Jian-chun Peng, Hui Jiang, and Yi-tao Liu. Deep learning based ensemble approach for probabilistic wind power forecasting. *Applied Energy*, 188:56–70, 2017. 1
- [95] Junke Wang, Yi Jiang, Zehuan Yuan, Binyue Peng, Zuxuan Wu, and Yu-Gang Jiang. OmniTokenizer: A Joint Image-Video Tokenizer for Visual Generation. *Advances in Neural Information Processing Systems*, 37:28281–28295, 2024. 2, 3
- [96] Wenjing Wang, Huan Yang, Zixi Tuo, Huiguo He, Junchen Zhu, Jianlong Fu, and Jiaying Liu. Swap Attention in Spatiotemporal Diffusions for Text-to-Video Generation, 2024. 3
- [97] Yunbo Wang, Mingsheng Long, Jianmin Wang, Zhifeng Gao, and Philip S Yu. Predrnn: Recurrent neural networks for predictive learning using spatiotemporal lstms. *Advances in neural information processing systems*, 30, 2017. 2, 7
- [98] D. S. Wilks. Indices of Rank Histogram Flatness and Their Sampling Properties. *Monthly Weather Review*, 147(2): 763–769, 2019. 8
- [99] Wang-Chun Woo. Application of Optical Flow Techniques to Rainfall Nowcasting. In *27th Conference on Severe Local Storms*. AMS, 2014. 2
- [100] Wang-chun Woo and Wai-kin Wong. Operational Application of Optical Flow Techniques to Radar-Based Rainfall Nowcasting. *Atmosphere*, 8(3):48, 2017. 2
- [101] Yuan-Kang Wu, Po-En Su, Ting-Yi Wu, Jing-Shan Hong, and Mohammad Yusri Hassan. Probabilistic Wind-Power Forecasting Using Weather Ensemble Models. *IEEE Transactions on Industry Applications*, 54(6):5609–5620, 2018. 1
- [102] Yazhou Xing, Yang Fei, Yingqing He, Jingye Chen, Jiaxin Xie, Xiaowei Chi, and Qifeng Chen. Large motion video autoencoding with cross-modal video vae, 2024. 2, 3
- [103] Nianzu Yang, Pandeng Li, Liming Zhao, Yang Li, Chen-Wei Xie, Yehui Tang, Xudong Lu, Zhihang Liu, Yun Zheng, Yu Liu, and Junchi Yan. Rethinking Video Tokenization: A Conditioned Diffusion-based Approach, 2025. 2, 3, 4, 6
- [104] Jingfeng Yao, Bin Yang, and Xinggang Wang. Reconstruction vs. generation: Taming optimization dilemma in latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. 5
- [105] Demin Yu, Xutao Li, Yunming Ye, Baoquan Zhang, Chuyao Luo, Kuai Dai, Rui Wang, and Xunlai Chen. Diff-Cast: A Unified Framework via Residual Diffusion for Precipitation Nowcasting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 27758–27767, 2024. 2, 3, 7
- [106] Lijun Yu, Jose Lezama, Nitesh Bharadwaj Gundavarapu, Luca Versari, Kihyuk Sohn, David Minnen, Yong Cheng,

- Agrim Gupta, Xiuye Gu, Alexander G. Hauptmann, Boqing Gong, Ming-Hsuan Yang, Irfan Essa, David A. Ross, and Lu Jiang. Language Model Beats Diffusion - Tokenizer is key to visual generation. In *The Twelfth International Conference on Learning Representations*, 2023. [3](#)
- [107] Qihang Yu, Mark Weber, Xueqing Deng, Xiaohui Shen, Daniel Cremers, and Liang-Chieh Chen. An Image is Worth 32 Tokens for Reconstruction and Generation, 2024. [3](#)
- [108] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *CVPR*, 2018. [2](#)
- [109] Yuchen Zhang, Mingsheng Long, Kaiyuan Chen, Lanxiang Xing, Ronghua Jin, Michael I Jordan, and Jianmin Wang. Skilful nowcasting of extreme precipitation with nowcast-net. *Nature*, 619(7970):526–532, 2023. [1](#), [2](#), [7](#)
- [110] Yitian Zhang, Long Mai, Aniruddha Mahapatra, David Bourgin, Yicong Hong, Jonah Casebeer, Feng Liu, and Yun Fu. Regen: Learning compact video embedding with (re-)generative decoder, 2025. [2](#), [3](#), [4](#), [6](#)
- [111] Long Zhao, Sanghyun Woo, Ziyu Wan, Yandong Li, Han Zhang, Boqing Gong, Hartwig Adam, Xuhui Jia, and Ting Liu. Epsilon-VAE: Denoising as Visual Decoding, 2025. [3](#), [4](#), [6](#)
- [112] Boyang Zheng, Nanye Ma, Shengbang Tong, and Saining Xie. Diffusion transformers with representation autoencoders, 2025. [2](#), [4](#), [5](#)